

A Lightweight, Offline Detector for Cassava Disease in Low-Resource West African Farms

Your Name*
Your Department, Your University
Your City, Your Country

Co-author Name
Their Department, Their University
Their City, Their Country

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WORKING TITLE — your 15-word promise

CropGuard: an offline, quantized detector for cassava disease on low-end phones in Guinea and Nigeria

Formulas that work:

1. *[Method]: [what it does] for [problem]* — e.g. “AgroNet: Lightweight Crop-Disease Detection for Offline Mobile Use”
2. *[Question or claim as a hook]* — e.g. “Do Multilingual Models Understand West African Languages?”
3. *[Resource] + [scale/context]* — e.g. “NaijaVoices: a 500-Hour Speech Corpus for Nigerian Languages”

Test: could a stranger guess your method, your problem AND your context from the title alone?

PICK ONE SCENARIO — or use your own research

A • AI for climate — flood forecasting, drought prediction, sparse-sensor spatiotemporal modelling in the Sahel.

B • Agriculture — pest and disease detection, food-crop yield optimisation, edge deployment on low-end phones.

C • Health — malaria diagnosis from microscopy, low-resource clinical NLP, triage under intermittent connectivity.

D • Local languages — MT and ASR for low-resource African languages, corpus construction, transfer learning under data scarcity.

ABSTRACT — three sentences, nothing more

S1 • CONTEXT + GAP *Where, who, how big is the stake — and what blocks the world today?*

In West Africa, cassava mosaic and brown-streak disease destroy up to 40% of yields for smallholder farmers, yet existing vision-based diagnostic models assume cloud connectivity and high-end hardware that rural farms do not have.

S2 • METHOD *Your one-line answer: what did you build, and what is new about it?*

We introduce CropGuard, a quantized and pruned convolutional detector that runs entirely offline on low-end Android devices, trained on a new dataset of 12,406 field images collected across 42 farms in Guinea and Nigeria and released under FAIR principles.

S3 • RESULTS + IMPACT *Named metrics, exact numbers, named baselines — then the door it opens.*

CropGuard reaches 78.3% top-1 accuracy while cutting inference latency by 40% relative to MobileNetV3, enabling farmers to diagnose disease in the field with no internet connection.

*Corresponding author: your.email@university.edu

Judge yourself with one question: could a stranger fund this project after reading these three sentences? If any sentence contains “very”, “novel”, “promising” or “revolutionary” without a number next to it, rewrite it.

Abstract

In West Africa, cassava mosaic and brown-streak disease destroy up to 40% of yields for smallholder farmers, yet existing vision-based diagnostic models assume cloud connectivity and high-end hardware that rural farms do not have. We introduce CropGuard, a quantized and pruned convolutional detector that runs entirely offline on low-end Android devices, trained on a new dataset of 12,406 field images collected across 42 farms in Guinea and Nigeria and released under FAIR principles. CropGuard reaches 78.3% top-1 accuracy while cutting inference latency by 40% relative to MobileNetV3, enabling farmers to diagnose disease in the field with no internet connection.

► **Coach:** The three sentences above are now, word for word, the skeleton of your Introduction. Section 1 shows you how to expand them into the funnel: context → gap → contribution.

1 Introduction

REVIEWER'S LENS — Introduction

What is the gap, and why is it urgent to solve it? If I cannot state your contribution in one sentence after reading this section, I stop reading.

The funnel — write it in this order:

1. **Context — the world.** Take sentence 1 of your abstract and give it stakes: who is affected, how many, with what economic or health cost. Cite a source for every number [3].
2. **Gap — what is missing.** Why do existing solutions fall short *for your setting*? Name them: “existing work requires X, which is unavailable in Y.”
3. **Contribution — you.** Take sentence 2 of your abstract.
4. **Teaser of the evidence.** Take sentence 3 of your abstract.

► **Coach:** Answer the Heilmeier Catechism here and your introduction writes itself: (i) What are you trying to do, in plain words with no jargon? (ii) How is it done today, and what are the limits? (iii) What is new in your approach, and why do you think it will succeed? (iv) Who cares — locally *and* globally?

► **Coach:** Never apologise for an African context. Frame it as a generalisable scientific challenge: “low-resource NLP for Wolof” becomes “robust transfer learning under severe data scarcity”; “drought forecasting in the Sahel” becomes “spatiotemporal forecasting in sparse-sensor, highly non-linear regimes”.

Our contributions.

- **A new method:** we propose ..., the first ... for low-resource settings;
- **A new resource:** we release a dataset of N samples collected in ..., with a dataset card and an open licence;
- **New evidence:** ... outperforms strong baselines by $Y\%$ on Z benchmarks, under real deployment constraints.

► **Coach:** Every claimed contribution must be verifiable in your Results section. Claim three things, prove three things.

2 Related Work

REVIEWER'S LENS — Related Work

How is this different from what already exists? Have you read my paper — and the three obvious ones you are competing against?

Position yourself in the conversation; do not write an annotated bibliography. Group prior work into two or three *stances*, and end each group with the limitation that motivates your work. For instance: transformer architectures [5] dominate the benchmark, but ...; and African-context studies [3, 4] show that ...

► **Coach:** Cite African research. If we do not cite each other, nobody else will.

3 Method

REVIEWER’S LENS — Method

Could a competent reader re-implement your system from this text alone? A method I cannot re-implement is a method I cannot trust — and cannot accept.

3.1 Notation

Keep notation consistent between this section, your pseudocode and your results tables, and define every symbol at first use: scalars are lowercase italic (x , y), vectors are lowercase bold ($\mathbf{x}$), matrices are uppercase bold ($\mathbf{X}$).

$$\hat{\mathbf{y}} = f_{\boldsymbol{\theta}}(\mathbf{X}), \quad \mathcal{L}(\boldsymbol{\theta}) = \frac{1}{N} \sum_{i=1}^N \ell(\hat{y}_i, y_i) \quad (1)$$

where $\boldsymbol{\theta}$ are the parameters of ..., N is ..., and $\ell(\cdot, \cdot)$ is ...

3.2 Data

Describe your dataset the way you would want someone else’s described to you, and publish a datasheet alongside the paper [1, 6].

DATASET CARD — your dataset deserves a full biography

Provenance: where, when, by whom, with what consent and ethics approval.
Size & composition: N samples, class balance, regions/languages covered.
Preprocessing: cleaning, filtering, augmentation — every transformation stated.
Splits & leakage control: how train/val/test were separated (by farm, by speaker...).
Known biases: urban vs rural coverage, device quality, seasonal effects.
Access & license: DOI (Zenodo), repository, and the terms others may reuse it under.

3.3 Architecture and training

Layers, losses, optimiser, learning-rate schedule, hardware. Put every hyperparameter in a table and state the search space you explored — not just the winning value.

3.4 Evaluation protocol

Metrics with formulas, train/val/test splits, the number of seeds, and the baselines you compare against — chosen fairly and named.

4 Experiments and Results

REVIEWER’S LENS — Results

What is the evidence — and is the comparison fair? Is the gain bigger than the noise?

► **Coach:** Note what Table 1 does: no vertical lines (booktabs), the best result is bold, every number has a unit, the caption is self-contained, and there is a *trade-off* column. On a \$50 phone, latency and parameter count are the contribution — report them.

► **Coach:** Before you claim “our model is better”: two accuracies differing is a hypothesis, not a result. Report a distribution (k -fold or ≥ 3 seeds, mean $\pm$ std). Test normality (Shapiro–Wilk); with small field datasets it usually fails, so use Kruskal–Wallis rather than ANOVA, then a post-hoc test (Nemenyi, Dunn) to say *which pair* differs, and report the critical difference.

Table 1: Top-1 accuracy (%) on the cassava test set, mean $\pm$ std over 3 seeds. Best result in **bold**. Latency measured on a \$50 Android device.

Model	Accuracy (%)	Latency (ms)	Params (M)
ResNet-50	71.2 $\pm$ 0.6	410	25.6
MobileNetV3	74.8 $\pm$ 0.4	180	5.4
EfficientNet-B0	76.1 $\pm$ 0.5	240	5.3
CropGuard (ours)	78.3 $\pm$ 0.3	108	3.1

5 Discussion

Interpret; do not repeat the table. Why does your method win? When does it not? What does it mean for the field and for deployment?

6 Limitations

REVIEWER’S LENS — Limitations

Where does this honestly stop working? If you hide the failure modes, I will find them — and that is how papers die in review.

Be honest, specific and self-critical. For example: “CropGuard was validated on three cassava varieties; performance under extreme low-light conditions and on 2G-only devices remains untested.”

► **Coach:** African-context limitations — connectivity, device constraints, seasonal data coverage — are scientifically valuable observations, not weaknesses. Own them, and turn each one into a line of future work.

7 Broader Impact and Ethics

Who benefits, who could be harmed, and what did you do about it? Consent for data collection, anonymisation of sensitive records, and fair recognition of the communities whose data you used.

8 Conclusion

One paragraph: restate the contribution, give your strongest number, and point to the future work that follows naturally from your limitations.

Reproducibility Statement

Code, data and trained models are available at <https://github.com/your/repo> (anonymised for review; archived with a Zenodo DOI at camera-ready). All library versions, random seeds and hardware are stated in Section 4.

► **Coach:** A reviewer who can run `scripts/run.sh` and reproduce your headline number is a reviewer who trusts you. Reproducibility is doubly strategic for African labs: it neutralises scepticism and showcases engineering quality.

Acknowledgements

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References

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